Notes: Bernanke and Gertler (AER, 1989)

Agency Costs, Net Worth, and Business Fluctuations

Contents

1 Big picture

- **Research question.** Can borrower balance sheets be an intrinsic source of business-cycle dynamics in an otherwise neoclassical real model?
- **Main idea.** When investment projects involve asymmetric information, external finance is costly. Higher borrower net worth reduces agency costs, so investment is easier to finance in good times and harder to finance in bad times.
- **Core mechanism.** A boom raises entrepreneurs' current income and net worth. Higher net worth improves collateralization, reduces costly auditing or monitoring, and increases real investment. The extra investment raises future capital and propagates the boom.
- **Reverse mechanism.** A downturn lowers entrepreneurial net worth, raises the external-finance wedge, and depresses investment. This lowers future capital and amplifies the downturn.
- **Modeling device.** The paper embeds costly state verification in an overlapping-generations real business-cycle environment. Entrepreneurs operate investment projects, lenders supply funds, and project outcomes are privately observed by entrepreneurs unless lenders audit at a real resource cost.
- **Benchmark contrast.** With perfect information, the model is deliberately rigged to have no interesting investment dynamics: investment and the capital stock are constant, and output moves only with the contemporaneous productivity shock.
- **Main result.** With agency costs, capital supply depends on entrepreneurial net worth. Productivity shocks and redistributions that affect borrower net worth therefore have persistent real effects.
- **Debt-deflation implication.** A redistribution from borrowers to lenders, such as through unindexed debt and unexpected deflation, lowers borrower net worth, raises agency costs, depresses investment, and can initiate a persistent downturn.
- **Economic takeaway.** Finance matters not only because it supplies funds, but because the distribution of net worth between borrowers and lenders determines how costly it is to convert saving into productive capital.

2 Incremental contribution of the paper

- **Formalizes the balance-sheet channel.** Earlier work argued that borrower solvency mattered in recessions and in the Great Depression. Bernanke and Gertler provide a formal general-equilibrium model in which borrower net worth drives investment dynamics.
- **Connects agency theory to business cycles.** The paper imports costly state verification from contract theory into a macroeconomic setting and derives external-finance costs from optimal contracting rather than assuming a reduced-form credit constraint.
- **Shows amplification without nominal rigidity.** Persistent investment and output fluctuations arise even though the model is neoclassical in spirit. The propagation mechanism is financial rather than Keynesian price stickiness.

- **Uses a clean benchmark.** The perfect-information version has no investment persistence by construction. This makes clear that the additional dynamics come from agency costs and borrower net worth.
- **Endogenizes capital supply.** Capital supply is no longer purely technological. It depends on the endogenous ability of entrepreneurs to finance projects without expensive verification.
- **Explains asymmetric fluctuations.** Positive shocks may have limited effects once borrowers are fully collateralized, while negative shocks can push borrowers into the constrained region. Thus investment downturns can be sharper than upturns.
- **Links distributional shocks to aggregates.** Redistributing wealth from entrepreneurs to lenders can reduce total investment even if aggregate resources are unchanged. Distribution matters because only entrepreneurs have direct access to the investment technology.
- **Prefigures the financial accelerator.** The paper is one of the foundational theoretical statements of the financial-accelerator logic: balance sheets affect external finance, and external finance feeds back into real activity.

3 Model setup

3.1 Agents and timing

The economy is an overlapping-generations model. Each generation lives for two periods. Young agents work and save; old agents consume returns from earlier saving.

- **Entrepreneurs.** A fraction η of each generation are entrepreneurs. They have direct access to investment projects and can transform current output into next-period capital.
- **Lenders.** The remaining agents are lenders. They save but do not have direct access to the investment technology.
- **Entrepreneur heterogeneity.** Entrepreneurs are indexed by $\omega \in [0, 1]$. Low-ω entrepreneurs are more efficient because they require less input to operate a project.
- **One-period financial contracts.** The model abstracts from long-term lending relationships. A period should be interpreted as the length of a typical financial contract.

Interpretation. The distinction between entrepreneurs and lenders is essential. A redistribution between them changes aggregate investment because only entrepreneurs can directly operate projects.

3.2 Goods and production

There are two goods: an output good and a capital good. Output can be consumed, stored, or invested in capital-producing projects.

Output is produced using capital and fixed labor:

$$y_t = \theta_t f(k_t),$$

where θ_t is an i.i.d. aggregate productivity shock and k_t is capital per head.

Investment projects are discrete and nondivisible. A project of entrepreneur type ω requires exactly $x(\omega)$ units of the output good, with $x'(\omega) > 0$. If undertaken, it produces random capital next period. The expected capital output per project is denoted κ.

- More efficient entrepreneurs have lower input cost $x(\omega)$.
- Project outcomes are idiosyncratic and independent across projects.
- There is no aggregate uncertainty in per capita capital output from idiosyncratic project risk.

Interpretation. Heterogeneity in $x(\omega)$ generates an upward-sloping supply of investment projects. Aggregate productivity affects current income and saving, not project quality directly.

3.3 Storage and savings

Output can also be stored directly. Storage yields gross return $r \geq 1$. Assumptions guarantee that aggregate saving exceeds profitable capital formation, so storage pins down the safe rate.

Entrepreneurs do not consume when young and are risk-neutral. Their average saving is simply

$$S_t^e = w_t L^e,$$

where w_t is the wage and L^e is the entrepreneurial labor endowment.

Lenders consume in both periods. Their saving is

$$S_t = w_t L - z_y^*(r),$$

where $z_y^*(r)$ is optimal young-age lender consumption.

Interpretation. The model creates a direct link from the current state of the economy to entrepreneurial saving. When productivity or inherited capital is high, wages and entrepreneurial net worth are high.

3.4 Costly state verification

Only entrepreneurs costlessly observe the realized return of their own projects. Lenders can learn the return only by auditing. Auditing uses γ units of capital.

- If information is perfect, $\gamma = 0$.
- If information is imperfect, $\gamma > 0$ and the financial contract may involve stochastic auditing.
- An entrepreneur who has more net worth can pledge more of the project input cost internally, so the incentive problem is less severe.

Interpretation. Agency costs are real resource costs arising from verification. The important comparative static is that expected agency costs fall as borrower net worth rises.

4 Perfect-information benchmark

4.1 Investment cutoff

With free auditing, information is perfect. Let $\hat{q}_{t+1}$ be the expected relative price of capital next period. An entrepreneur of type ω invests if expected investment surplus is nonnegative:

$$\hat{q}_{t+1}\kappa - rx(\omega) \geq 0.$$

The cutoff entrepreneur $\bar{\omega}$ satisfies

$$\hat{q}_{t+1}\kappa - rx(\bar{\omega}) = 0.$$

All entrepreneurs with $\omega \leq \bar{\omega}$ invest.

4.2 Capital supply

Since ω is uniformly distributed, the number of projects undertaken per capita is

$$i_t = \bar{\omega}\eta,$$

and next-period capital is

$$k_{t+1} = \kappa i_t.$$

Combining these relationships gives the perfect-information capital supply curve:

$$\hat{q}_{t+1} = \frac{r\,x(k_{t+1}/\kappa\eta)}{\kappa} \qquad [SS].$$

4.3 Capital demand

Capital demand is determined by the condition that the expected price of capital equals its expected marginal product:

$$\hat{q}_{t+1} = \theta f'(k_{t+1}) \qquad [DD].$$

The demand curve is downward-sloping in $(k_{t+1}, \hat{q}_{t+1})$ space because the marginal product of capital is higher when capital is scarce.

4.4 Why the benchmark has no dynamics

The SS and DD curves do not depend on k_t or on the current productivity realization θ_t. Thus their intersection determines a constant k_{t+1} every period.

- Investment is fixed.
- Capital is fixed.
- Output varies only because of the current productivity shock.
- There is no propagation of shocks through investment.

Economic message: The benchmark is intentionally sterile. Any subsequent persistence in investment and output must come from the agency-cost channel.

5 Asymmetric information and optimal contracting

5.1 Contracting problem

When $\gamma > 0$, lenders cannot observe project outcomes without paying an auditing cost. A financial contract must specify transfers and auditing probabilities so that entrepreneurs truthfully report project outcomes.

The contract trades off two forces:

- **Incentive compatibility.** Entrepreneurs must not want to report a low outcome when the true outcome is high.
- **Verification cost.** Auditing deters misreporting but destroys resources.

The appendix shows that expected auditing costs are nonincreasing in entrepreneur collateral S^e. When borrower net worth is high enough, the lender can be repaid even in the worst state, and no auditing is needed.

5.2 Full collateralization threshold

For each entrepreneur type ω, define $S^*(\omega)$ as the level of entrepreneurial saving that makes the project fully collateralized:

$$S^*(\omega) = x(\omega) - \frac{\hat{q}}{r}\kappa_1.$$

If the entrepreneur contributes at least $S^*(\omega)$, the project can be financed without costly auditing.

Interpretation. This is the key balance-sheet statistic. The closer an entrepreneur is to $S^*(\omega)$, the lower the external-finance wedge.

5.3 Good, fair, and poor entrepreneurs

The asymmetric-information case divides entrepreneurs into three economic types.

- **Good entrepreneurs.** These entrepreneurs are efficient enough to undertake projects even when they face positive expected auditing costs. They invest and put all saving into their project until full collateralization.
- **Fair entrepreneurs.** These entrepreneurs have positive-surplus projects under full collateralization, but not enough net worth to make all of them financeable. Their investment is effectively rationed by internal equity.
- **Poor entrepreneurs.** These entrepreneurs are too inefficient to invest. Storage dominates the project even after considering financing arrangements.

Economic message: Agency costs do not merely raise a uniform cost of capital. They change which projects can be financed and make marginal projects especially dependent on borrower net worth.

5.4 The role of entrepreneurial net worth

Higher S^e affects investment through two margins.

- **Intensive margin among good entrepreneurs.** The same good projects are undertaken with lower expected auditing costs.
- **Extensive margin among fair entrepreneurs.** More fair entrepreneurs can become fully collateralized and operate projects.

Both margins increase effective capital supply. This is why the distribution of wealth between borrowers and lenders matters for aggregate investment.

6 Within-period equilibrium with agency costs

6.1 Audit probability and constrained project entry

For good entrepreneurs, let $p(\omega)$ be the probability of auditing in the low-output state. In the paper's two-outcome version,

$$p(\omega) = \max\left\{\frac{rx(\omega) - \hat{q}\kappa_1 - rS^e}{\hat{q}[\pi_2(\kappa_2 - \kappa_1) - \pi_1\gamma]}, 0\right\}.$$

This probability is decreasing in $\hat{q}$ and in S^e.

For fair entrepreneurs, let $g(\omega)$ be the fraction that can invest. The model writes

$$g(\omega) = \min\left\{\frac{rS^e}{rx(\omega) - \hat{q}\kappa_1}, 1\right\}.$$

This fraction is increasing in $\hat{q}$ and in S^e.

Interpretation. A higher capital price or higher borrower net worth relaxes the constraint. More projects can be financed with less verification.

6.2 Agency-cost capital supply

Total capital formation under asymmetric information combines the net capital output of good entrepreneurs and the capital output of fair entrepreneurs that can invest. The capital supply curve is

$$k_{t+1} = \left\{\kappa\bar{\omega} - \left[\int_0^{\underline{\omega}} \pi_1\gamma p(\omega)d\omega + \int_{\underline{\omega}}^{\bar{\omega}} \kappa(1 - g(\omega))d\omega\right]\right\}\eta.$$

The bracketed terms are the deadweight losses from auditing and the lost capital formation from excluded fair entrepreneurs.

6.3 Properties of the agency-cost supply curve

The agency-cost capital supply curve $S'S'$ has three important properties.

- **It lies left of the perfect-information supply curve.** For a given expected capital price, less capital is supplied because verification destroys capital and some positive-surplus projects cannot be financed.
- **It is upward-sloping.** Higher expected capital prices make more projects financeable and reduce auditing needs.
- **Its position depends on S^e.** Higher entrepreneurial saving shifts $S'S'$ down and to the right, toward the perfect-information SS curve. Lower entrepreneurial saving shifts it up and to the left.

Economic message: Capital supply is no longer a purely technological relationship. It is state-dependent because financial frictions depend on borrower net worth.

6.4 Comparative statics

- **Positive income shock.** A higher current productivity realization or inherited capital stock raises wages and entrepreneurial saving. Higher S^e lowers agency costs, shifts $S'S'$ right, raises k_{t+1}, and lowers $\hat{q}_{t+1}$.
- **Debt-deflation shock.** A redistribution of endowment from entrepreneurs to lenders reduces S^e even if aggregate resources are unchanged. The $S'S'$ curve shifts left, k_{t+1} falls, and the expected capital price rises.
- **Perfect-information contrast.** These comparative statics do not operate when information is perfect because capital supply is independent of entrepreneurial wealth.

7 Aggregate dynamics

7.1 Propagation of productivity shocks

With agency costs, a productivity shock affects current wages and thus entrepreneurial net worth. Because net worth affects capital supply, the shock changes future capital formation.

A positive temporary productivity shock has the following sequence:

$$\theta_t \uparrow \quad \Rightarrow \quad w_t \uparrow \quad \Rightarrow \quad S_t^e \uparrow \quad \Rightarrow \quad \text{agency costs} \downarrow \quad \Rightarrow \quad k_{t+1} \uparrow .$$

The higher future capital stock raises future income and can keep investment higher in later periods.

A negative productivity shock works in reverse:

$$\theta_t \downarrow \quad \Rightarrow \quad S_t^e \downarrow \quad \Rightarrow \quad \text{agency costs} \uparrow \quad \Rightarrow \quad k_{t+1} \downarrow .$$

Economic message: This is an accelerator mechanism. Output affects investment not only through desired capital, but through borrower balance sheets and the cost of external finance.

7.2 Asymmetric dynamics

The dynamic effects of shocks may be asymmetric. If the economy starts at the perfect-information capital stock and entrepreneurs are fully collateralized, further positive shocks may not matter much for investment because agency constraints are already slack. But negative shocks can push borrowers back into the constrained region.

Interpretation. Bad shocks can generate larger and more persistent real responses than good shocks. The financial constraint often binds on the downside.

7.3 Debt-deflation dynamics

A debt-deflation transfers wealth from debtors to creditors when nominal debt is unindexed and the price level unexpectedly falls. In the model, the analogue is a redistribution of endowment from entrepreneurs to lenders.

- Entrepreneurial saving S^e falls.
- The agency-cost supply curve shifts up and left.
- Current investment falls.
- The lower capital stock depresses future output and future entrepreneurial net worth.
- The effect can persist even though aggregate resources are initially unchanged.

Economic message: Balance-sheet shocks can initiate fluctuations. They do not merely amplify productivity shocks.

8 Figures and extracted exhibits

8.1 Figure 1 and Figure 3: Capital-market equilibrium

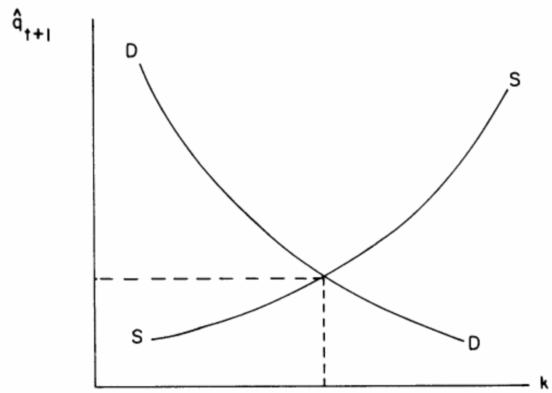

FIGURE 1

Figure 1: Perfect-information capital supply and demand. The upward-sloping SS curve is capital supply and the downward-sloping DD curve is capital demand. Their intersection pins down a constant next-period capital stock.

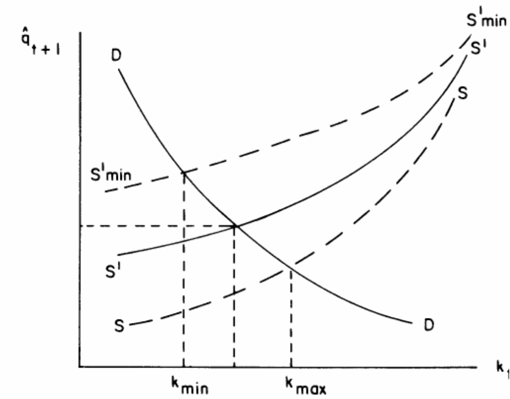

FIGURE 3

Figure 3: Agency-cost capital supply. The imperfect-information supply curve $S'S'$ lies left of SS and shifts with entrepreneurial saving S^e. Lower borrower net worth shifts capital supply left and lowers investment.

8.2 Figure 2 and capital-supply equations

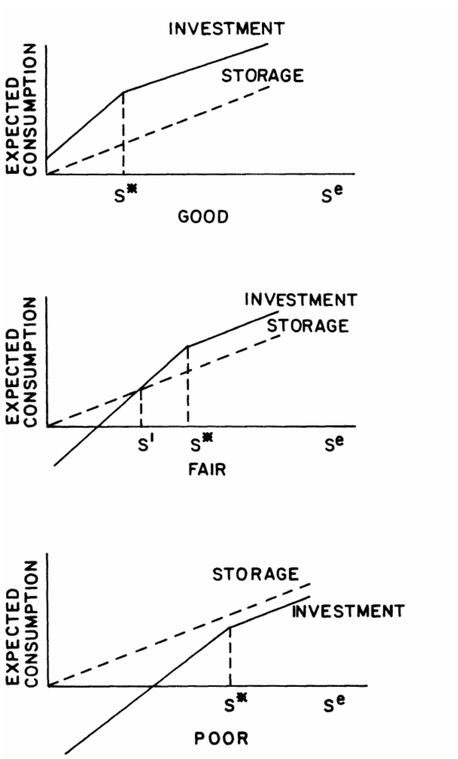

FIGURE 2

Figure 2: Opportunity sets for good, fair, and poor entrepreneurs. Good entrepreneurs invest; poor entrepreneurs store; fair entrepreneurs are constrained by collateralization and internal net worth.

trepreneurial saving in t are determined.

We would like to know the supply and demand curves for capital. Consider the determination of capital supply, for a given expected relative price of capital, $\hat{q}$.[26] For $\omega \le \underline{\omega}$, define $p(\omega)$ to be the probability that an entrepreneur of type ω is audited (in the bad state). The function $p(\omega)$ is defined by

$$(27) \quad p(\omega) = \max\left(\frac{rx(\omega) - \hat{q}\kappa_1 - rS^e}{\hat{q}(\pi_2(\kappa_2 - \kappa_1) - \pi_1\gamma}, 0\right)$$

for $\omega \le \underline{\omega}$ (compare (22)). $p(\omega)$ is decreasing in $\hat{q}$ and in S^e; $p(\omega) = 0$ for $S^e \ge S^*(\omega)$.

Fair entrepreneurs (with types between $\underline{\omega}$ and $\bar{\omega}$), because of the "collateralization lottery," do not face the agency cost of auditing when they invest; but only the fraction of fair entrepreneurs who win the lottery are able to invest. Let $g(\omega)$, defined for $\underline{\omega} < \omega \le \bar{\omega}$, be the fraction of fair entrepreneurs of type ω who can invest (and $1 - g(\omega)$ be the fraction who are excluded). Using the fact that $g(\omega) = S^e/S^*(\omega)$, and substituting from (26), we have

$$(28) \quad g(\omega) = \min\left(\frac{rS^e}{rx(\omega) - \hat{q}\kappa_1}, 1\right)$$

Total capital formation (per head) in this case is given by

$$(29) \quad k_{t+1} = \left[\kappa\underline{\omega} - \pi_1\gamma\int_0^{\underline{\omega}} p(\omega)\, d\omega\right]\eta + \left[\kappa\int_{\underline{\omega}}^{\bar{\omega}} g(\omega)\, d\omega\right]\eta,$$

where the expression in the first set of brackets reflects capital formation (net of auditing costs) by good entrepreneurs, and the second expression in brackets is capital formation by fair entrepreneurs. (29) can be rewritten as

$$(30) \quad k_{t+1} = \left\{\kappa\bar{\omega} - \left[\int_0^{\underline{\omega}} \pi_1\gamma p(\omega)\, d\omega + \int_{\underline{\omega}}^{\bar{\omega}} \kappa(1 - g(\omega)\, d\omega\right]\right\}\eta \quad [SS].$$

(30) is the capital supply curve for the $\gamma > 0$ case. It is depicted in Figure 3 as the $S'S'$ curve, along with the perfect informa-

Extracted equations: Audit probability $p(\omega)$, investment fraction $g(\omega)$, and aggregate capital formation under agency costs. These equations formalize how borrower net worth affects capital supply.

8.3 Appendix contracting problem

from the beginning.) We allow a general stochastic auditing strategy: The lenders can commit in advance to auditing an announcement of outcome i with probability p_i. The total input cost of the project is x (here we hold the entrepreneur's "efficiency," ω, fixed). The entrepreneur's contribution is his savings S^e, and the interest rate is r, so that the lenders' total required return is $r(x - S^e)$. The entrepreneur's (borrower's) formal problem is

$$\text{(A1)} \qquad \max_{\{c_i^a, c_i, p_i\}} \sum_{i=1}^{n} \pi_i \left(p_i c_i^a + (1 - p_i) c_i \right)$$

subject to

$$\text{(A2)} \qquad \sum_{i=1}^{n} \pi_i \left[\hat{q}\kappa_i - \left(p_i c_i^a + (1 - p_i) c_i \right) - \hat{q} p_i \gamma \right] \geq r(x - S^e) \qquad (\lambda_1)$$

$$\text{(A3)} \qquad p_i c_i^a + (1 - p_i) c_i \geq (1 - p_j)\left(c_j + \hat{q}(\kappa_i - \kappa_j) \right) \qquad i = 2, \ldots, n \quad j < i \qquad (\lambda_{2ij})$$

$$\text{(A4)} \qquad c_i \geq 0 \qquad i = 1, 2, \ldots, n \qquad (\lambda_{3i})$$

$$\text{(A5)} \qquad c_i^a \geq 0 \qquad i = 1, 2, \ldots, n \qquad (\lambda_{4i})$$

$$\text{(A6)} \qquad p_i \geq 0 \qquad i = 1, 2, \ldots, n \qquad (\lambda_{5i})$$

$$\text{(A7)} \qquad 1 \geq p_i \qquad i = 1, 2, \ldots, n \qquad (\lambda_{6i})$$

Extracted appendix problem: The stochastic-auditing contract maximizes entrepreneur expected consumption subject to lender participation, truth-telling, nonnegativity, and audit-probability constraints.

9 How to read the paper's logic

9.1 Step 1: Remove financial frictions

The perfect-information model is a control case. If auditing is free, all positive-surplus projects can be financed. Borrower wealth does not matter because lenders can observe outcomes and enforce repayment without deadweight costs.

Read this as: the real technology alone does not create cyclical persistence in this setup.

9.2 Step 2: Add costly verification

Once outcomes are privately observed, lenders must use contracts that sometimes audit. Auditing is costly, so external finance creates a wedge between the social value of a project and the private returns that can be pledged to lenders.

Read this as: financing is not neutral. The Modigliani–Miller logic fails because the financial contract consumes real resources and excludes some projects.

9.3 Step 3: Link agency costs to net worth

The more an entrepreneur can contribute internally, the less she needs to promise to outside lenders. The incentive problem becomes easier, and expected auditing costs fall.

Read this as: net worth is not just a passive state variable. It directly changes the efficiency of investment finance.

9.4 Step 4: Link net worth to the cycle

Entrepreneurial net worth is procyclical because wages and income are procyclical. Therefore agency costs are countercyclical.

Read this as: the economy has an internal propagation mechanism: good times make finance easier, and easy finance helps good times continue.

10 Interpretation and limitations

10.1 What the model identifies

The model identifies a theoretical mechanism by which borrower balance sheets can amplify and propagate real shocks. It is not an empirical estimation and does not quantify the channel.

- The mechanism is qualitative and structural.
- The central comparative static is robust: agency costs fall with borrower net worth.
- The business-cycle implication is that financial distress raises the cost of external finance and depresses investment.

10.2 Why risk neutrality matters

Risk neutrality simplifies the contracting problem and focuses attention on agency costs rather than risk sharing. It also permits the paper to interpret net worth as collateral that relaxes incentive constraints.

Limitation: In richer models, risk aversion, diversification, and hedging could alter the contract and the strength of the net-worth channel.

10.3 Why one-period contracts matter

The model abstracts from repeated lending relationships. Long-term contracts could allow lenders to use future profits and reputation to relax current constraints.

Limitation: Borrower net worth in a multi-period model should include not only current assets but also the secure part of expected future profits.

10.4 Why the model has no aggregate demand channel

The paper focuses on the supply side of investment finance. Debt-deflation lowers investment through borrower net worth rather than through deflationary aggregate demand.

Limitation: The model by itself does not explain a stock market crash or demand-side collapse without additional ingredients.

10.5 Policy interpretation

The paper does not provide a full welfare or policy analysis. Still, it suggests why policies that affect borrower balance sheets may have aggregate consequences.

- Transfers to borrowers can raise investment when agency constraints bind.
- Transfers to lenders can depress investment even if total resources are unchanged.
- Stabilizing borrower net worth may reduce amplification of downturns.

11 Short summary

- The paper builds a neoclassical business-cycle model with costly state verification.
- Entrepreneurs have direct access to investment projects; lenders provide funds but cannot observe project outcomes without costly auditing.
- Borrower net worth reduces agency costs because more internal funding makes contracts easier to enforce.
- With perfect information, investment is constant and output has no persistence beyond the productivity shock.
- With agency costs, capital supply depends on entrepreneurial saving.
- Productivity shocks affect investment because they change entrepreneurial wages and net worth.
- Positive shocks improve balance sheets, reduce agency costs, and raise investment.
- Negative shocks weaken balance sheets, raise agency costs, and lower investment.
- The model therefore generates a financial accelerator: balance-sheet movements amplify and propagate real shocks.
- A debt-deflation-like redistribution from entrepreneurs to lenders lowers investment even when aggregate resources do not fall.
- The key insight is that the allocation of net worth across agents matters for aggregate capital formation.

12 Conclusion

12.1 Core conclusion

Bernanke and Gertler show that borrower balance sheets can be a source of business-cycle dynamics. When external finance is costly because of asymmetric information, borrower net worth determines the agency cost of financing investment. Since borrower net worth is procyclical, agency costs are countercyclical, and investment becomes more volatile and persistent than in a frictionless real model.

12.2 Economic intuition

The model's intuition is simple: firms with strong balance sheets can finance investment more cheaply because they need less outside monitoring. Firms with weak balance sheets must rely more on external finance, which is costly when lenders cannot verify outcomes. Thus good times make finance easier and bad times make finance harder.

12.3 Debt-deflation insight

The paper also explains why redistributions can have real effects. Moving wealth from borrowers to lenders lowers investment because lenders do not have the same direct access to productive projects. This gives a formal channel through which debt-deflation can depress real activity.

12.4 Broader importance

The paper's enduring contribution is to turn financial distress into a macroeconomic state variable. Borrower net worth is not merely a symptom of the cycle; it is part of the propagation mechanism. This logic became central to later work on the financial accelerator and on credit-channel theories of monetary and macroeconomic fluctuations.

12.5 Takeaway

Finance amplifies cycles because net worth governs access to external funds. Strong balance sheets lower agency costs and support investment; weak balance sheets raise agency costs and restrict investment. The same real technology can therefore produce very different macroeconomic dynamics depending on borrowers' financial condition.